

Multiple Choice

1. **Answer:** C

Options: A) It is used to call procedures with addresses that are farther than 2^{16} bytes away.; B) It cannot return a value.; C) It cannot pass parameters by reference.; D) It cannot call procedures implemented in a different language.

Note: Correct option: C - It cannot pass parameters by reference.

2. **Answer:** B

Options: A) `sort('ab')`; B) `sorted('ab')`; C) `"ab".sort()`; D) `1/0`

Note: Correct option: B - `sorted('ab')`

3. **Answer:** C

Options: A) `{1}`; B) `{1,2,2,3,4}`; C) `{1,2,3,4}`; D) `{4,3,2,2,1}`

Note: Correct option: C - `{1,2,3,4}`

4. **Answer:** A

Options: A) `xyz`; B) `xy`; C) `xxzy`; D) `xxxxy`

Note: Correct option: A - `xyz`

5. **Answer:** A

Options: A) Reference counting is well suited for reclaiming cyclic structures.; B) Reference counting incurs additional space overhead for each memory cell.; C) Reference counting is an alternative to mark-and-sweep garbage collection.; D) Reference counting need not keep track of which cells point to other cells.

Note: Correct option: A - Reference counting is well suited for reclaiming cyclic structures.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It can never be proven.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def is_Perfect_Square(n) : | return True | return False`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def parallelogram_perimeter(b,h): | return perimeter`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key